



Green Synthesis of Graphene Oxide Quantum Dots: Biosensor for Dopamine Detection

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Abstract

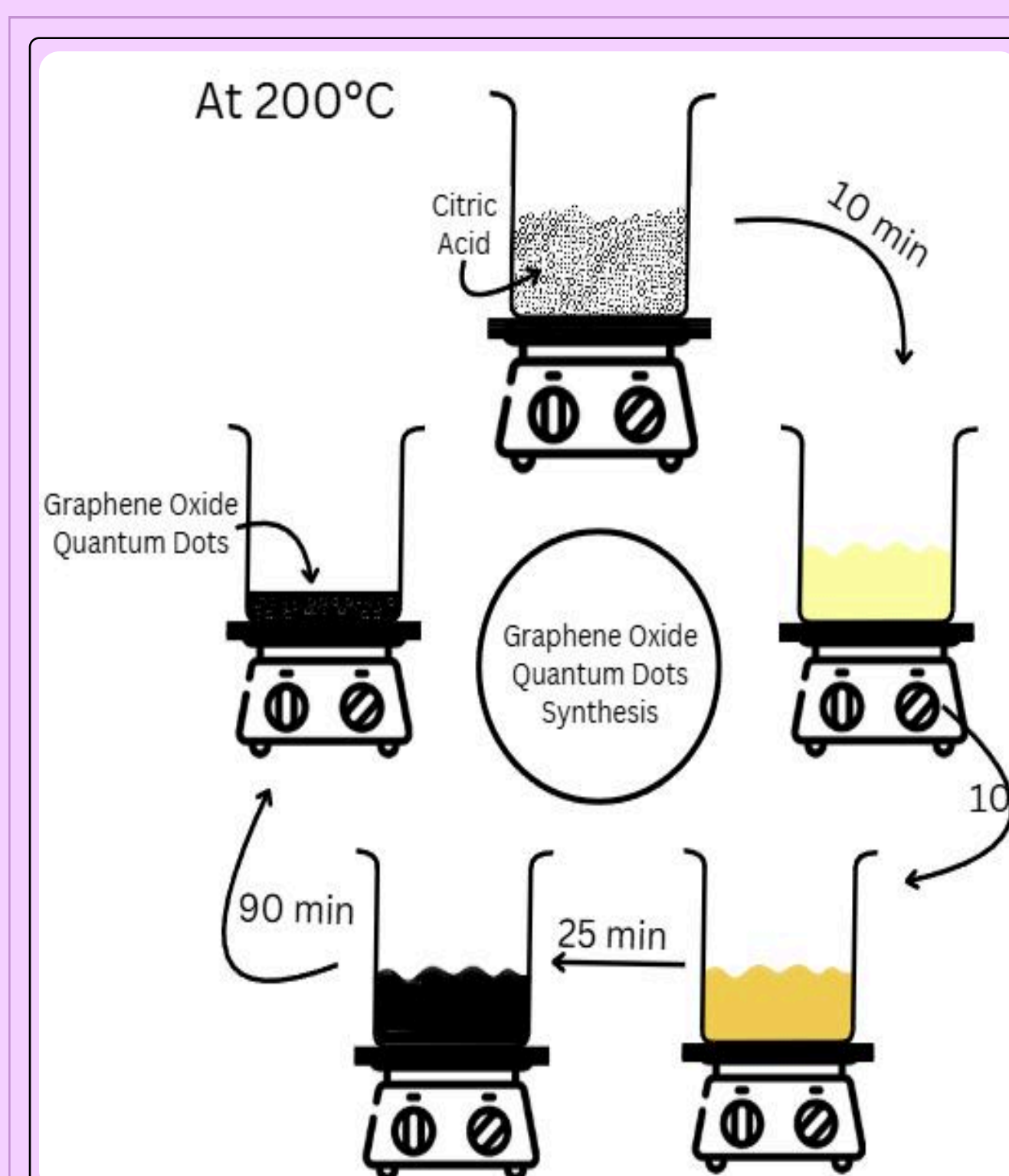
Nanotechnology has significantly enhanced the sensitivity and performance of biosensors, with graphene oxide quantum dots (GO QDs) emerging as promising materials due to their high surface area, tunable chemistry, and quantum size effects. This study presents a sustainable, green synthesis approach for producing graphene oxide (GO) QDs through the pyrolysis of citric acid, eliminating the need for hazardous, highly concentrated acids, and enhancing the overall sustainability of GO QD production. Synthesized GO QDs were characterized using UV-visible spectroscopy, fluorescence spectroscopy, Fourier-transform infrared spectroscopy (FT-IR), X-ray diffraction (XRD), and dynamic light scattering (DLS). UV-visible spectroscopy revealed a primary absorption peak at 230 nm with a shoulder peak at 300 nm, which confirmed the synthesis and presence of oxygen-containing functional groups in the GO QDs. FT-IR spectrum showed characteristic peaks at 3340 cm^{-1} (O-H stretching vibrations), 1730 cm^{-1} and 1630 cm^{-1} (C=O stretching of carbonyl groups), indicating the functionalization of GO QDs with oxygen-containing groups. XRD analysis revealed the crystalline nature of particle with an average GO QDs particle size of 7.46 nm, while DLS measurements indicated a hydrodynamic size of 23.4 nm, consistent with the quantum dot nature of the material. Further, the interaction of GO QDs with dopamine (Dop), a vital biomarker for neurodegenerative diseases, was explored in detail. UV-visible titration studies demonstrated a high binding constant of $3.65 \times 10^8 \text{ M}^{-1}$, indicating a strong interaction between dopamine and GO QDs. Fluorescence quenching experiments revealed a detection limit (LOD) of 0.21 μM , suggesting GO QDs high sensitivity and broad detection capabilities. Electrochemical analysis using cyclic voltammetry (CV) further validated the biosensing capability of GO QDs. Redox peaks corresponding to Dopamine concentrations (100-500 nM) were observed, with a highly linear response. Notably, the CV-based detection limit was found to be 44.64 nM, which suggests its high sensitivity and reliability at lower concentration. Furthermore, the results revealed the GO QDs based sensor showed strong binding and excellent selectivity towards dopamine, underscoring its potential as a robust biosensor for precise and efficient biomarker detection.

Methodology

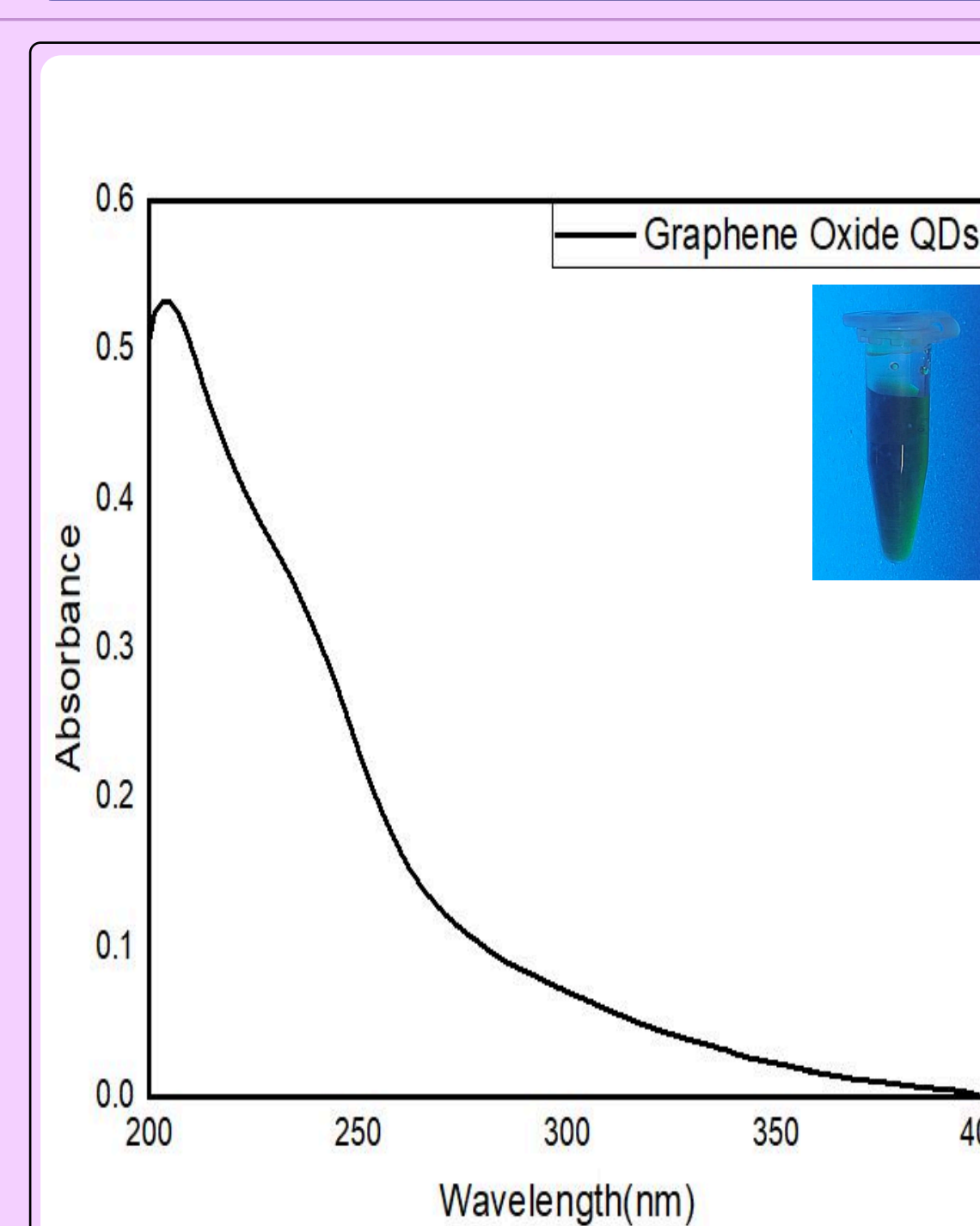
Graphene oxide quantum dots (GO QDs) were synthesized using a green thermal decomposition method with citric acid as the precursor. For characterization, the synthesized GO QDs were analyzed using several techniques. UV-Visible spectroscopy was performed to confirm the optical properties and successful synthesis. Fourier-transform infrared spectroscopy was utilized to identify the functional groups present in the GO QD structure. X-ray diffraction analysis was conducted to determine the crystalline structure and interlayer spacing of the GO QD particles. Additionally, Dynamic Light Scattering (DLS) was employed to assess the size distribution and zeta potential of the GO QD particles in suspension, providing insights into their colloidal stability. To investigate the interaction between GO QDs and dopamine, various analytical methods were employed. UV titration involved preparing dopamine solutions of varying concentrations and incrementally adding them to the GO QD solution, followed by recording the UV-Vis spectra to evaluate binding affinities. Fluorescence quenching studies were conducted by measuring the changes in fluorescence intensity of the dopamine-GO mixture upon dopamine addition, allowing for the determination of quenching constants and interaction mechanisms. Cyclic voltammetry (CV) was utilized to examine the electrochemical behavior of dopamine in the presence of GO QDs, comparing the redox profiles with and without dopamine to assess changes in electron transfer properties.

Results & Discussion

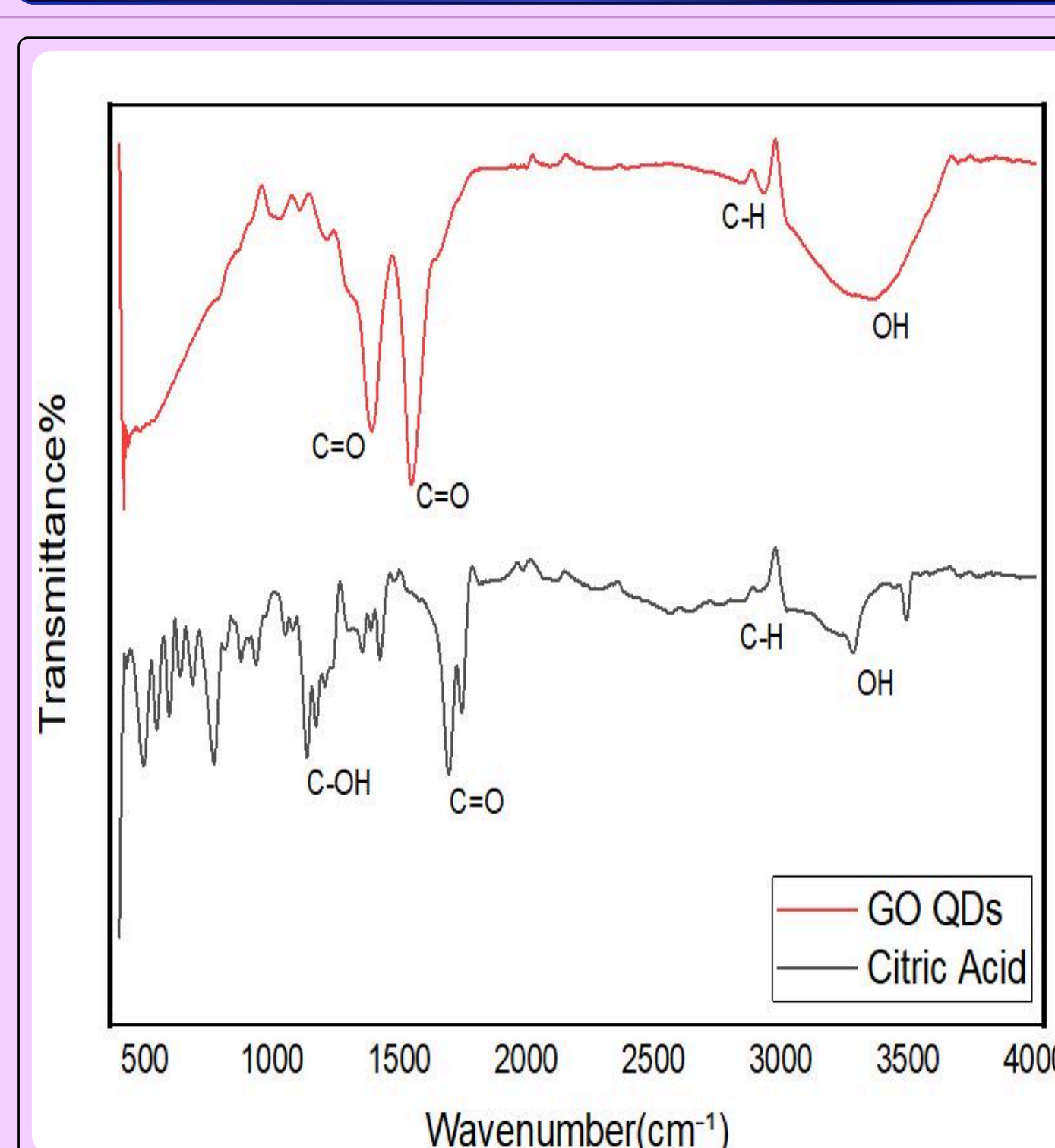
Synthesis Protocol



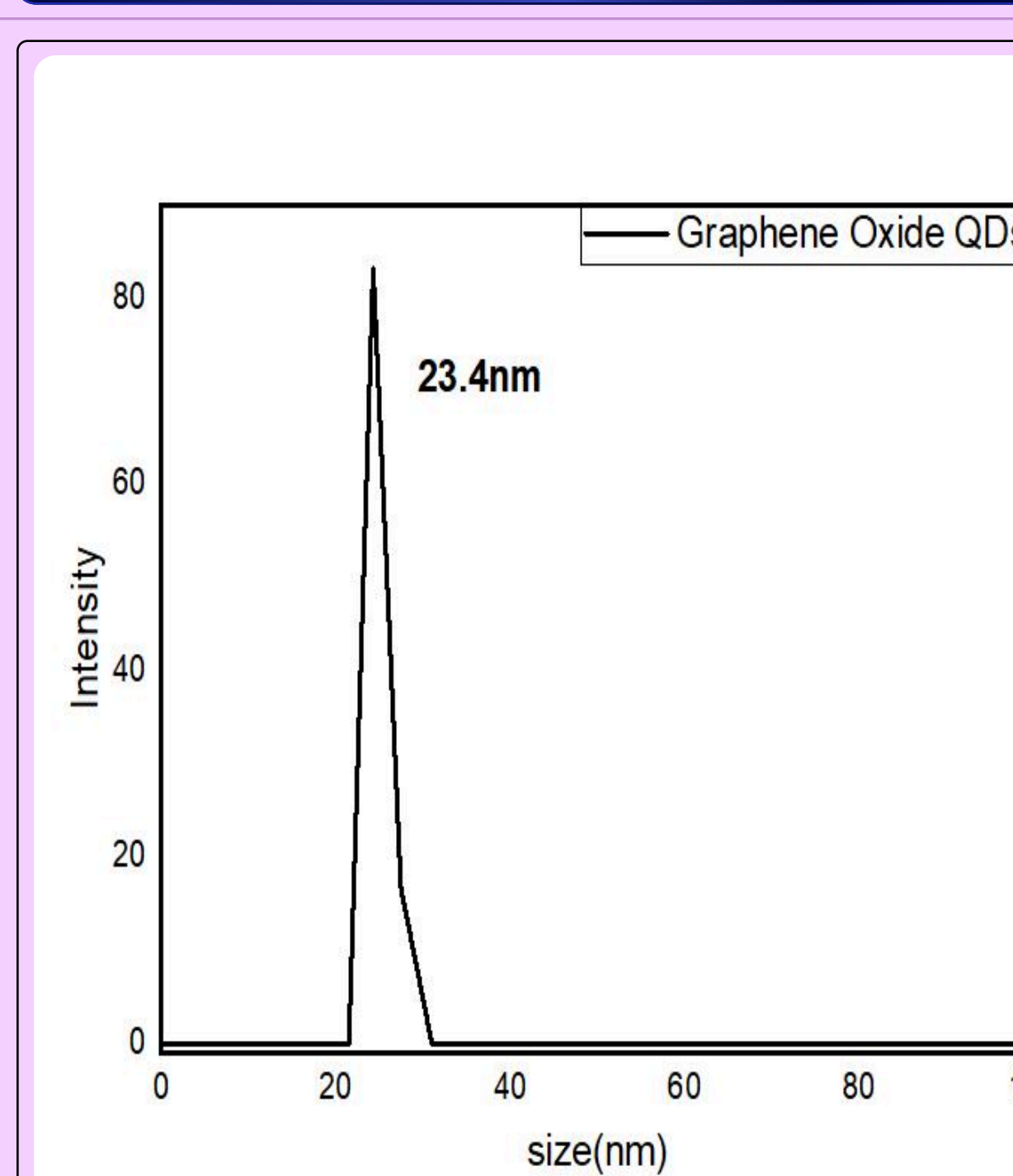
UV-Vis Spectroscopy



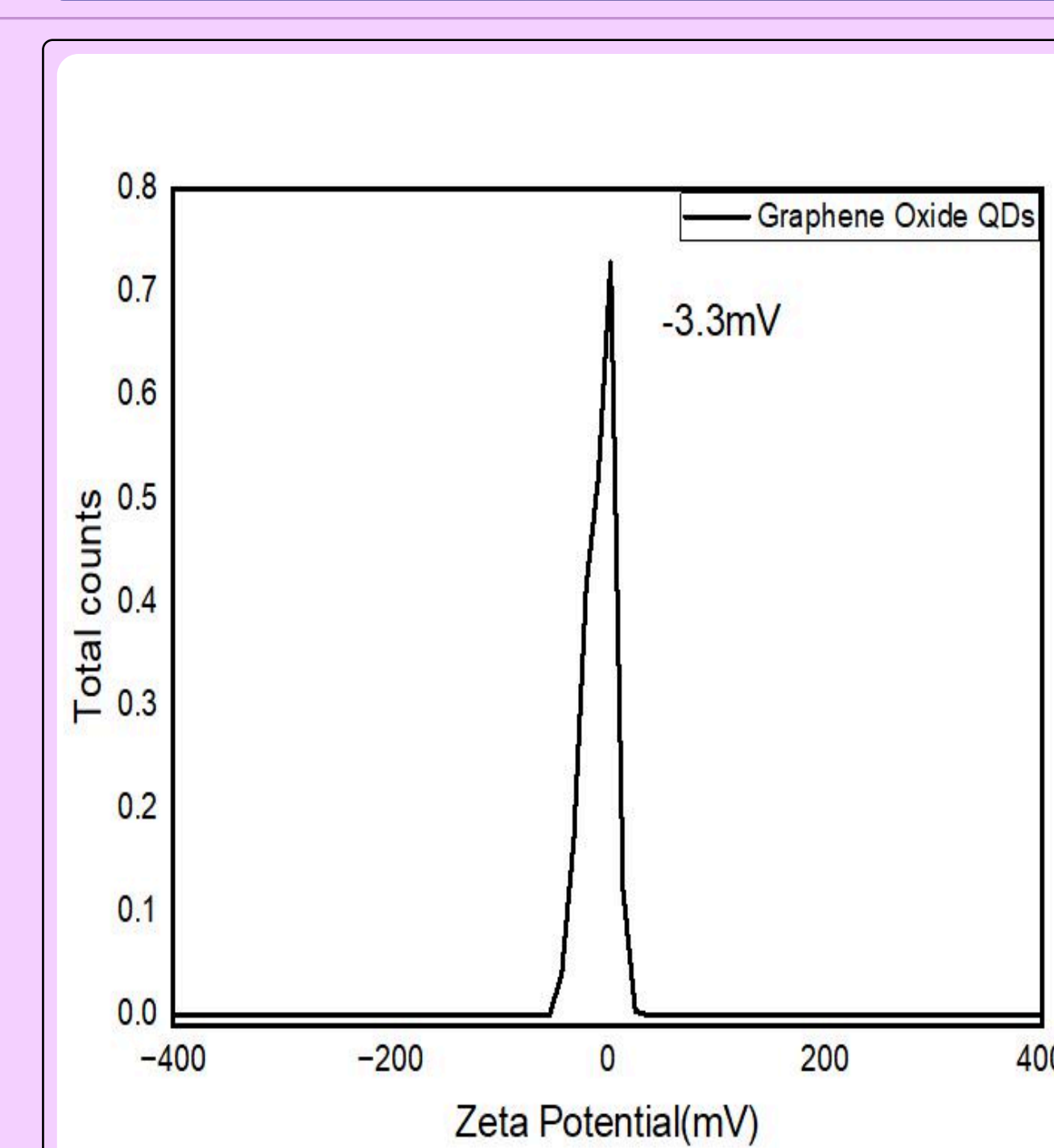
FT-IR Studies



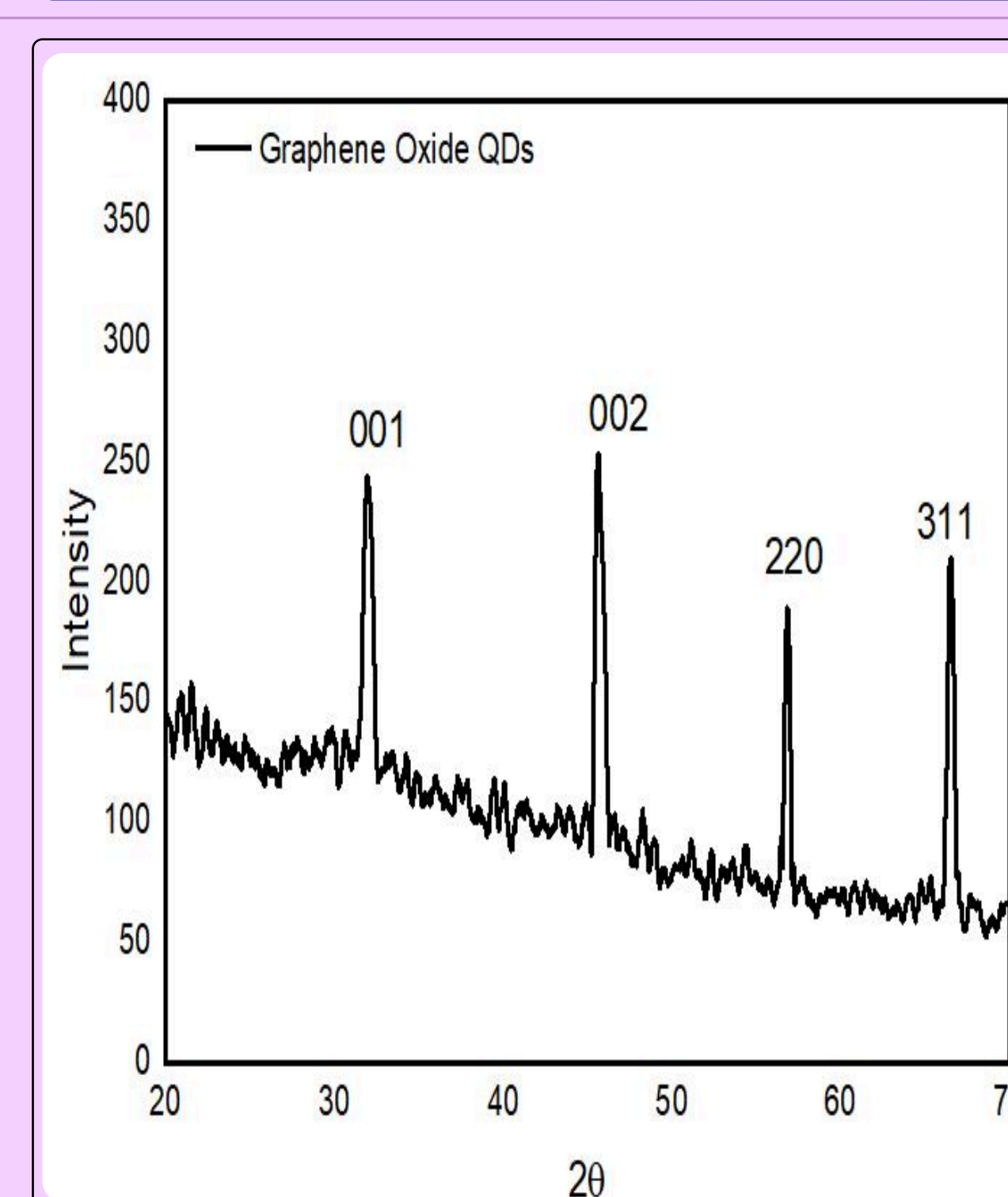
Dynamic Light Scattering



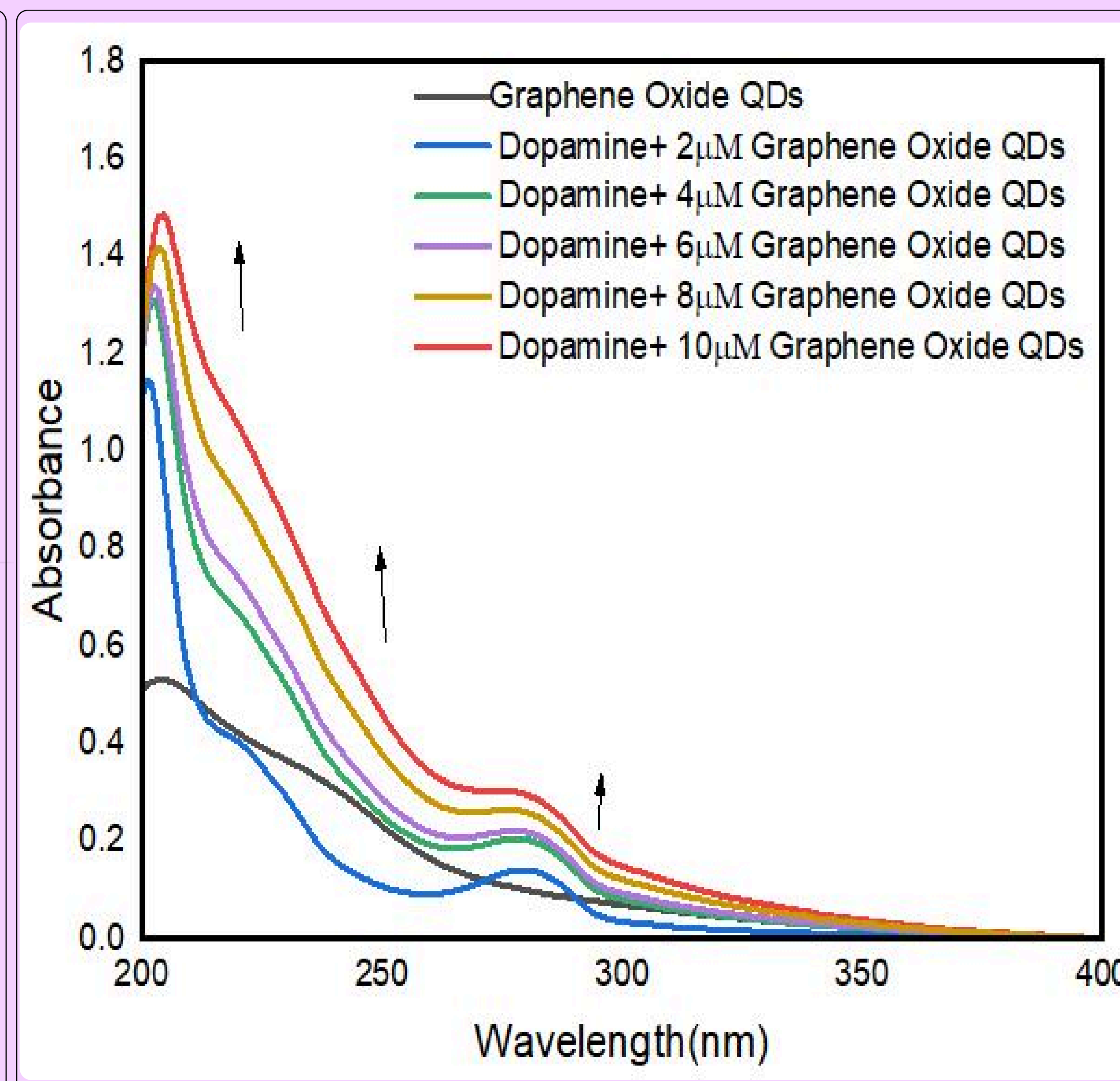
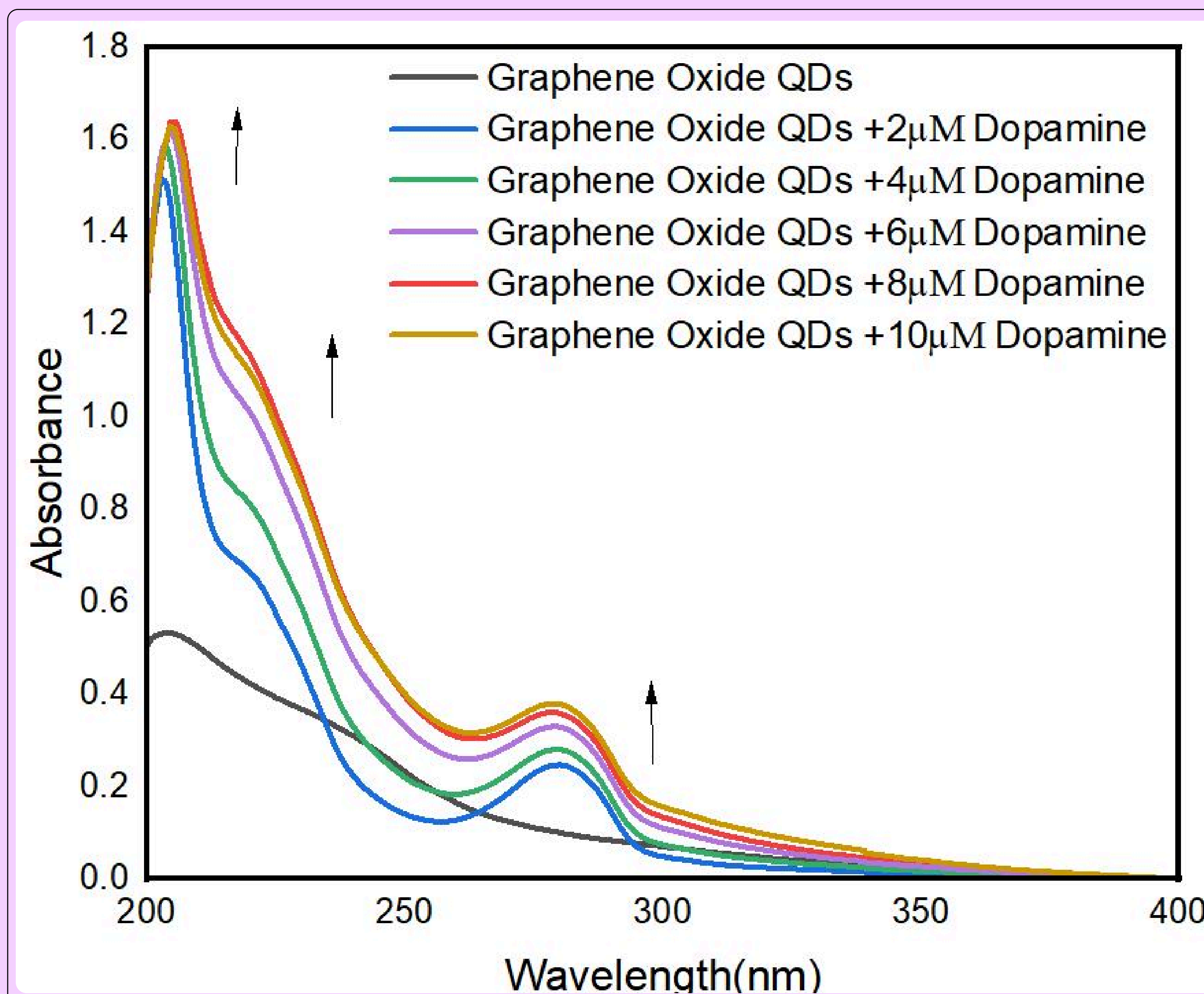
Zeta Potential



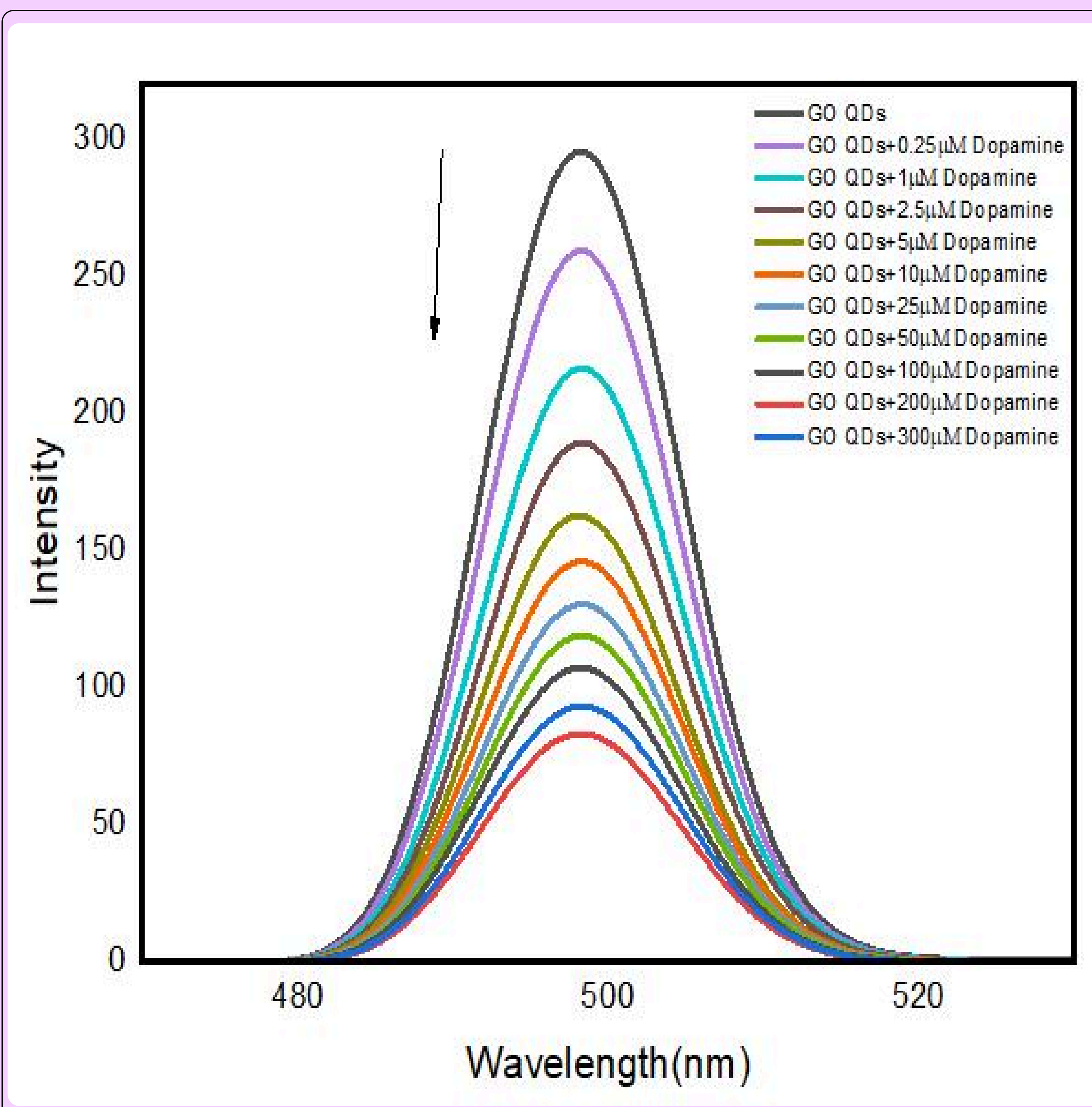
X-Ray Diffraction



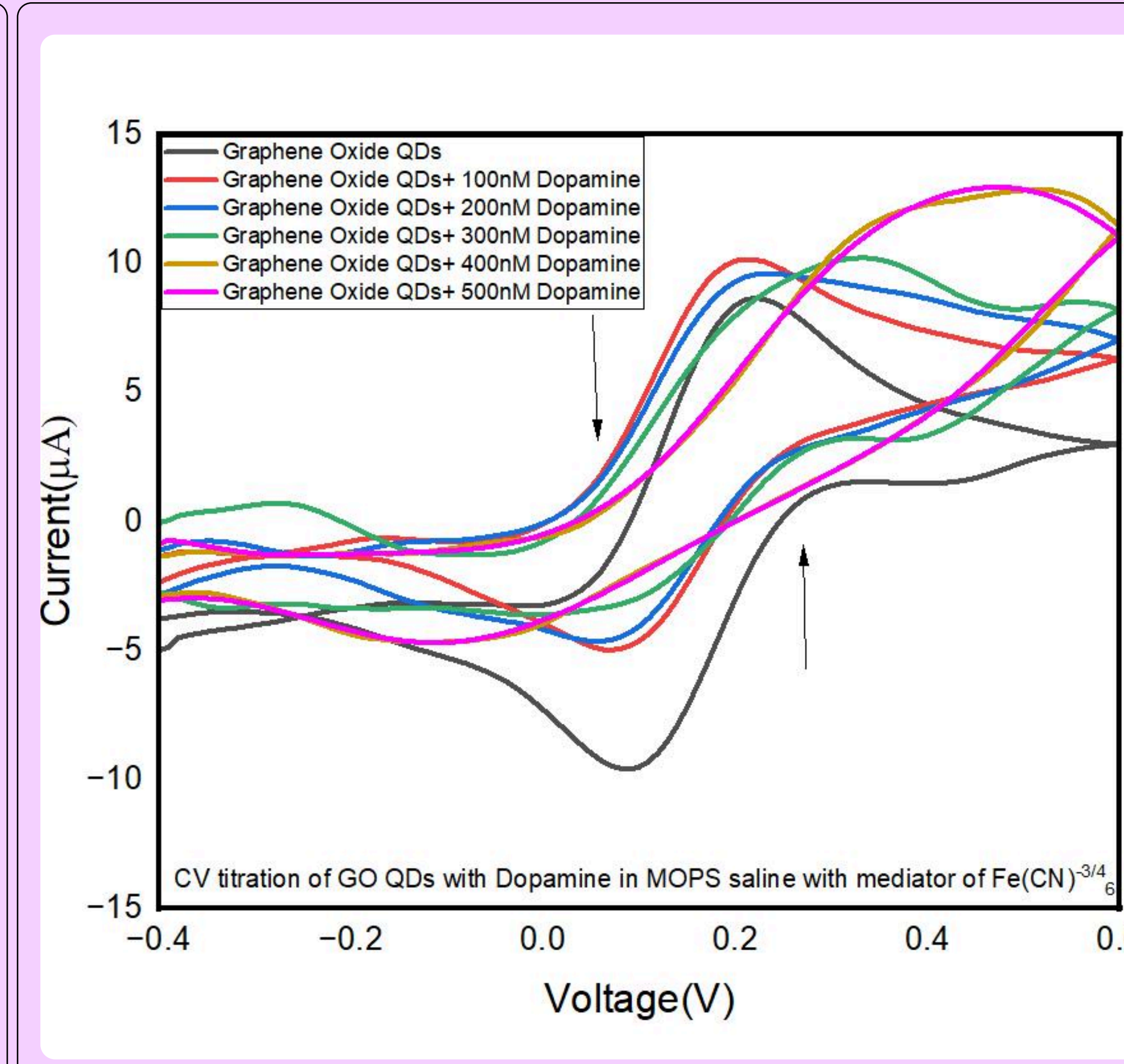
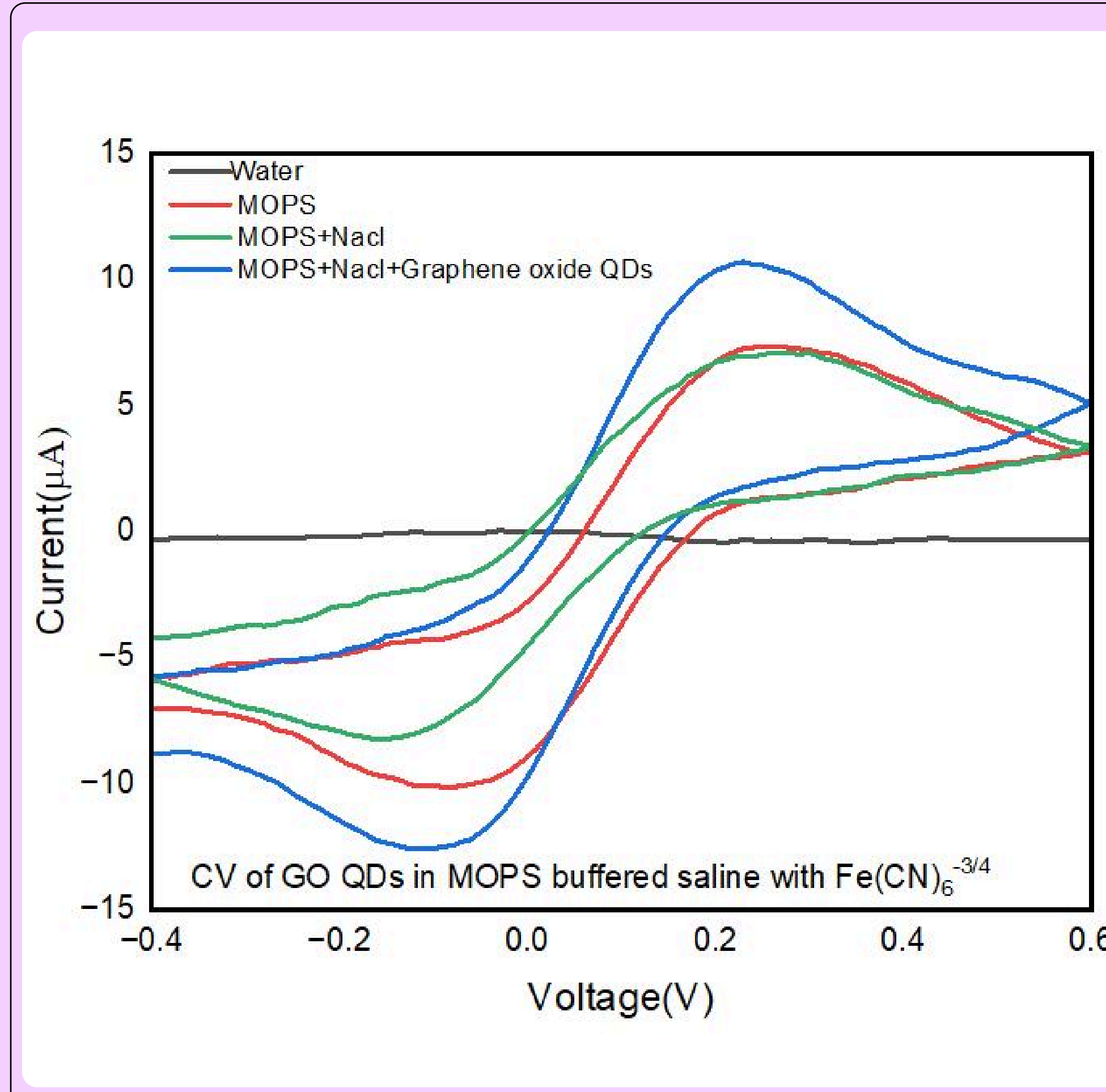
UV-Visible Titration Studies



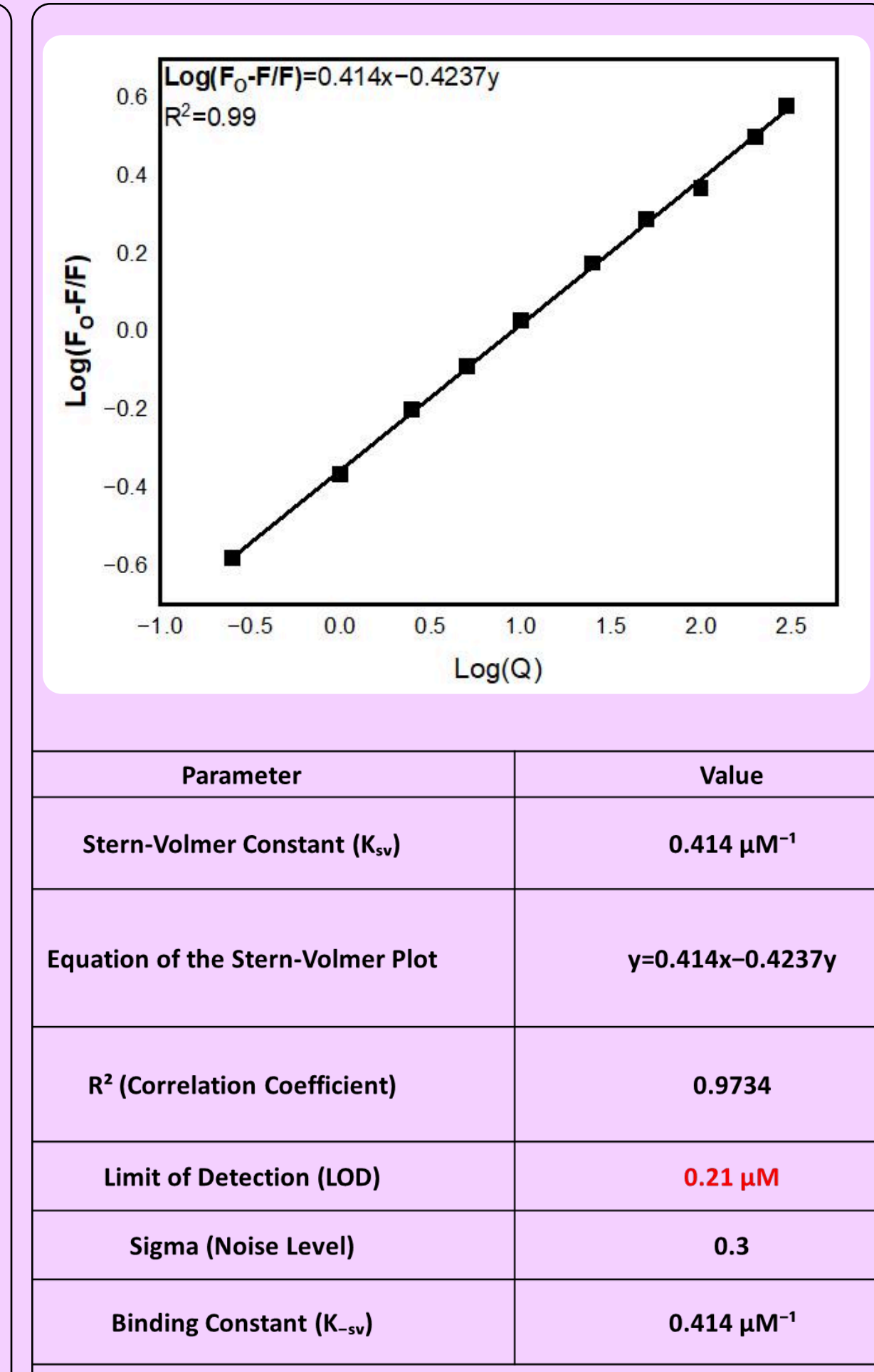
Fluorescence Studies



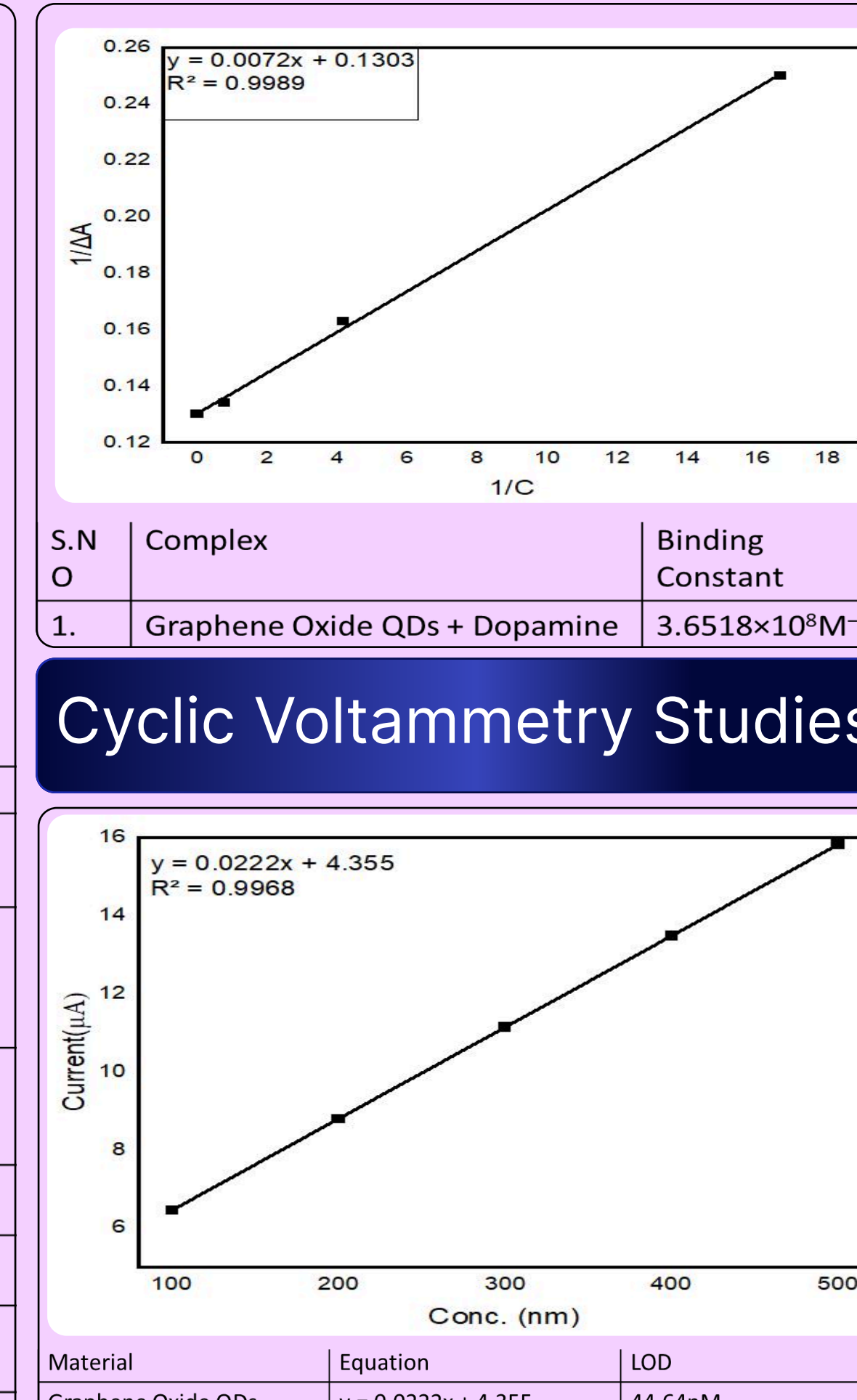
Cyclic Voltammetry Studies



Stern Volmer Studies



Binding Studies



Comparison with other Studies

Comparing the detection performance of different fluorescent method for Dopamine detection:			
Materials	Linear Range	LOD	Reference
NGQDs - GP73 _{apt} - MoS ₂ @RGO nanosheets	1.5–5.9 μg/mL	4.54 ng/ml	(Liang et al., 2023)
Au@N-GQD	0–100 μM	0.43 μM	(Das et al., 2019)
Graphene oxide	0.25 to 20 nM	94 nM	(Chen et al., 2011)
NSGQD	0 to 42 nM	7.8 nM	(Duhan & Ojari, 2023)
GO QDs	0.25 to 300 μM	0.21 μM	This work

Comparing the detection performance of different electrochemical methods for Dopamine detection:			
Materials	Linear Range	LOD	Reference
TiO ₂ /G ₂ C ₃ N ₄	10–90 μM	0.14 μM	(Balkourani et al., 2024)
rGO/PPy	0.2 mM and 1.2 mM	0.061 μM	(Akin et al., 2023)
g-C ₃ N ₄ /MWNTs/GO	2–100	0.22	(Wang et al., 2022)
GQD@MWNTs/GCE	0.25–250	0.095	(Arumugasamy et al., 2020)
N-GO	0–15 μM	45 nM	(Aukštakajytė et al., 2024)
rGO/M/GCE	5–42	0.11	(Gaidukevicius et al., 2022)
GO QDs	100–500nM	44.64nM	This work

Conclusions

- Successfully synthesized graphene oxide quantum dots using a sustainable green thermal decomposition method with citric acid, eliminating the need for hazardous chemicals.
- Characterization techniques (UV-Vis, FTIR, XRD, DLS) validated the formation, structural integrity, and size of 23.4nm distribution of the GO QDs, ensuring their suitability for biomedical applications.
- Demonstrated a high binding constant ($3.65 \times 10^8 \text{ M}^{-1}$) between GO QDs and dopamine, indicating a robust and stable interaction essential for effective biomarker detection.
- Achieved low detection limits through fluorescence quenching of 0.21 μM and cyclic voltammetry of 44.64 nM, showcasing the GO QDs' exceptional sensitivity for dopamine quantification.
- The GO QD based sensor exhibited excellent selectivity and reliability, underscoring its potential as a robust biosensor for precise and efficient detection of dopamine and other neurodegenerative biomarkers.

References

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